What Does Discussion Teach? Behavioral and Mechanistic Effects of Multi-Agent RLVR at Matched Compute

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Abstract

Multi-agent discussion is a popular addition to RLVR training pipelines, but it's rarely separated from the extra compute the discussion phase consumes. I train three matched Qwen3-4B models on the same math problems: a solo baseline (S), a discussion-trained model (D) that exchanges drafts with a peer on disagreement, and a compute-matched control (C) that spends the same extra compute on additional solo rollouts. C and D both outperform S on hard out-of-distribution evals and are close to each other on accuracy; on the most behaviorally distinctive metric, D's adversarial flip rate (the rate of flipping from correct to wrong under a wrong counter-argument) is roughly $6\times$ lower than S's and $4\times$ lower than C's. I then characterize three residual-stream axes (commitment, reflection, peer-framing) across all four models and find a unified pattern: D's typical reasoning state sits at base-like engagement on every axis, while solo training pulls typical state along these axes. Steering experiments confirm the reflection axis is causally responsible for D's distinctive surface behavior, and a smoke experiment shows that suppressing this axis at counter-argument time pushes D's flip rate above unsteered S's. The commitment direction, extracted contrastively on rollout-level correct vs wrong labels and originally labeled a "correctness direction," turns out under steering to control commitment behavior (format compliance, revision rate) rather than correctness. This is a methodological note on what CAA on rollout-level correctness labels actually picks up when the within-problem diagnostic is infeasible. Together these results characterize what discussion teaches: not a continuous shift in reasoning state, but sharper event-time discrimination on the reflection axis.

1 Introduction

Multi-agent discussion has become a popular addition to RLVR training. Systems like CRISP [Baumgartner and Agrawal, 2026], MAPoRL [Park et al., 2025], and Dr. MAS [Feng et al., 2026] report accuracy gains from peer interaction. But every one of these systems adds compute when discussion fires (extra forward passes, answer comparison, draft exchange) without isolating whether the gain comes from the interaction itself or just from spending more compute on hard problems.

The question this paper asks: *at matched compute, what does discussion actually teach a model that solo training does not?*

I train three Qwen3-4B-Instruct models from the same base via GRPO on DAPO-17K, with identical compute budgets. Model S trains solo. Model D trains with a two-player protocol (Alice T=0.6, Bob T=1.0) that triggers discussion on disagreement. Model C is the critical control: same two players, same disagreement detection, but on disagreement each player generates extra solo rollouts instead of exchanging drafts. C and D consume the same total compute; the only thing that differs is whether the extra compute is spent on *discussion* or on *more solo sampling*.

Behaviorally, C and D are close on hard out-of-distribution evals, both beating S; on easier tiers all three are roughly tied. The accuracy story alone is "discussion is no worse than C, neither is much better than S." But D has a distinctive fingerprint: its adversarial flip rate is roughly $6\times$ lower than S's and $4\times$ lower than C's, and it produces about half as many surface reflection markers per rollout at similar pass@1.

The bulk of this paper is mechanism. I extract three contrastive directions per condition. The commitment direction takes mean(correct rollout) minus mean(wrong rollout) at the last token before `\boxed{`. The reflection direction takes mean(predecessor of reflection-trigger) minus mean(matched non-trigger token). The peer-framing direction takes mean(peer-framed prompt) minus mean(source-unspecified prompt) at byte-identical suffix positions. I then characterize how the trained conditions differ from base on each axis. The cross-condition geometry, mid-layer robustness checks, typical-engagement projections, and steering experiments together produce one finding: D's training preserves base-like typical reasoning state across all three axes, while sharpening discrimination at event moments. Solo training, in contrast, pulls typical state along these axes during default reasoning.

Two methodological points fall out. First, the "correctness direction" extracted via CAA on rollout-level labels causally controls commitment behavior (format compliance, revision rate), not correctness itself; the pass@1 correlation in the contrastive labels is a downstream artifact. Second, the alternative hypothesis that D's adversarial robustness is peer-sycophancy rather than genuine resistance is directly falsified by the typical-engagement analysis on the peer-framing axis.

Contributions.

- A controlled three-condition comparison isolating discussion from extra compute, with the matched-compute control disentangling protocol from sampling.
- A multi-axis mechanistic characterization showing that D's training preserves base-like typical reasoning state on every axis measured.
- Causal validation of the reflection axis via activation steering, including a specificity control and a cross-model transfer check, plus a preliminary result linking the reflection axis to adversarial robustness.
- A methodological observation that CAA on rollout-level correctness labels captures answer-commitment rather than correctness when the within-problem diagnostic is infeasible.

2 Setup

Training conditions. All three models are Qwen3-4B-Instruct-2507 trained via GRPO [Guo et al., 2025] on DAPO-17K [Yu et al., 2025] math problems. Identical optimizer, learning rate, and total compute. The only variable is what happens during the rollout phase on disagreement problems.

Model S (Solo) trains with 8 standard rollouts per problem. Standard single-agent RLVR.

Model D (Discussion) trains with two players (Alice at T=0.6, Bob at T=1.0). When the two players reach different majority votes, they exchange representative rollouts and generate revised solutions. GRPO updates Alice's weights from the post-discussion rollouts. A persuader bonus of $\gamma = 0.3$ is added to the correct pre-discussion rollouts of a player whose answer their peer adopted after discussion. This follows the CRISP discussion-only design [Baumgartner and Agrawal, 2026].

Model C (Compute Control) runs the same two-player setup but on disagreement each player generates one additional rollout from a self-draft prompt structured identically to D's discussion prompt, except the two solutions shown are both from the same player. C consumes the same compute as D per disagreement event; the only difference is whether the extra compute is spent on peer interaction or on more samples from the same distribution.

Training data and hyperparameters. All three conditions train on DAPO-Math-17K [Yu et al., 2025] filtered to a curriculum of 2,023 problems with base solve rate in $[0.25, 0.75]$ at T=0.6

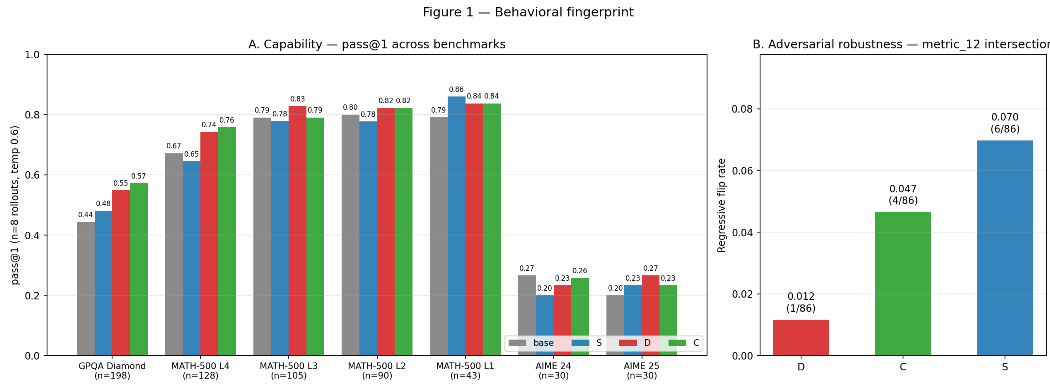

Figure 1: Behavioral fingerprint. **Panel A:** pass@1 across benchmarks. C and D beat S on the two largest harder-than-L3 evals (GPQA Diamond, MATH-500 L4). L1-L3 essentially tied across conditions. **Panel B:** regressive flip rate (rate of flipping from correct to wrong under a plausible wrong counter-argument) on the 86-problem intersection. D's resistance is the most behaviorally distinctive feature in the data: 0.012 (D) vs 0.047 (C) vs 0.070 (S).

(10K-problem random scan). GRPO with DCPO-style asymmetric clipping (low/high $\epsilon = 0.2/0.28$), peak learning rate 10^{-6} with 20-iter linear warmup and cosine decay to $0.1\times$ peak by iter 200, forward Jensen-Shannon KL against frozen base with coefficient 0.005, 8 rollouts/player/problem, 8 problems/batch. 200 iterations on $2\times$ H100. "iter 100" denotes the milestone checkpoint analyzed throughout this paper. Full hyperparameter table and compute-matching protocol details in Appendix A.

Evaluation. pass@1 measured at T=0.6, 8 rollouts per problem, on MATH-500 [Hendrycks et al., 2021], GPQA Diamond [Rein et al., 2024], and AIME 2024/25. Adversarial robustness uses an 86-problem intersection set where all three trained conditions solve $\geq 5/8$ rollouts unsteered; we append plausible wrong counter-arguments after correct answers and measure how often the model flips.

All probing experiments use the same four models. We extract residual-stream activations at layers $\{0, 4, 8, 12, 16, 20, 24, 28, 32, 36\}$ of the 36-layer Qwen3-4B and apply contrastive analysis per the methodology of Rimsky et al. [2024].

3 Behavioral fingerprint

Capability ordering on hard evaluations. Figure 1A shows pass@1 across seven evaluations. On GPQA Diamond (n=198), C 0.573 $>$ D 0.549 $>$ S 0.480 $>$ base 0.444. On MATH-500 L4 (n=128), C 0.758 $>$ D 0.742 $>$ base 0.672 $>$ S 0.646. The C $\geq$ D $>$ S pattern holds on the two largest harder-than-L3 evals; L1-L3 is essentially flat across conditions; AIME 2024/25 is too small (n=30 each) to be informative.

Discussion does not beat its compute-matched control on accuracy. C is 1-3pp ahead of D on GPQA and MATH-500 L4. On accuracy alone, the picture is "spending the extra compute on discussion is roughly as effective as spending it on more samples, with maybe a small edge to more samples."

Adversarial robustness is where D is distinctive. Figure 1B shows the regressive flip rate on the 86-problem intersection set. D flips 1/86 (0.012). C flips 4/86 (0.047). S flips 6/86 (0.070). D's resistance is more than $4\times$ C's and roughly $6\times$ S's. This gap is the most behaviorally distinctive feature in the data and is the thing the mechanistic analysis has to explain.

Reflection markers. D produces meaningfully fewer surface reflection markers per rollout ("wait", "actually", "let me check") than S or C at similar pass@1. Combined with the adversarial result, this raises a natural question: is D more committed because it reflects more (internally, without verbalizing) or because it reflects less? The mechanistic analysis in §4 answers this directly.

A note on the discussion training signal. Cumulative analysis of D's training disagreement cases (n=111) shows that productive asymmetric persuasion (one player correct, the other incorrect, and the correct one prevails) accounts for only about 15% of disagreement outcomes. Mutual swap, where both players flip, accounts for roughly 47%. The persuader bonus that D's reward is supposed to provide is heavily diluted by the symmetry of the two players; both are sampled from the same checkpoint and frequently move toward each other's answers, canceling out the productive signal.

This is the most likely explanation for why D does not beat C on accuracy: the discussion training signal is largely symmetric, so the gradient information per disagreement event is weaker than the matched compute spent on more solo rollouts. Breaking this symmetry — e.g. training D with a heterogeneous Bob from a different model or checkpoint — is the natural next step and is what I am working on this coming week. Whether a heterogeneous-Bob version of D ("D-prime") translates the persuader bonus into actual accuracy gains over C is the question for the follow-up.

4 Three representational axes

The behavioral results give an under-determined picture: D is not the most accurate, but it has a distinctive robustness profile. The mechanistic question is whether D's training reshapes the model's internal representations in ways that explain that profile. I extract three contrastive directions per condition, each measuring a different aspect of how the model represents reasoning state.

Commitment axis. Mean(correct rollout) − mean(wrong rollout) of residual-stream activations at the last reasoning token before `\boxed{` opens. Originally labeled "correctness direction" by analogy to Zhang et al. [2025]; the steering experiments in §5 show this direction controls answer-commitment behavior (whether the model emits a boxed answer at all) rather than correctness, motivating the relabeling.

Reflection axis. Mean(predecessor of reflection-trigger token) − mean(matched non-trigger token from the same rollout). Trigger words: "wait", "actually", "let me check/reconsider/verify", "hmm", "hold on", "I think I made". Probing the predecessor rather than the trigger token itself eliminates surface-form leakage. Matched-pair sampling controls for problem identity and rollout length. Follows the contrastive design of Zhu et al. [2025].

Peer-framing axis. Mean(peer-framed counter-argument prompt) − mean(source-unspecified prompt) at shared-suffix positions of paired prompts that differ only in their framing intro ("Your peer Bob says..." vs "Consider the following..."). By construction the boundary tokens are byte-identical between variants, so the layer-0 direction is the zero vector: a hard sanity check that passed in every condition.

All three directions pass the standard contrastive-direction sanity gates: layer-0 leakage bounded, random-label z-scores exceed 5 (reflection and peer-framing are well over 50), bootstrap CIs exclude zero at peak. Peak layer is L32 for the commitment and peer-framing axes; L36 for the reflection axis.

4.1 Cross-condition geometry holds across layers

For each direction I re-ran cross-condition cosine analysis at multiple late layers (commitment: L24/28/32; reflection: L24/28/36; peer-framing: all 10 probed layers). The question is whether the cross-condition pattern depends on a specific layer choice or holds across the late residual stream. Figure 2 shows the result.

Commitment axis. D-to-base cosine is layer-invariant at $\approx +0.44$ (L24/28/32: 0.439, 0.445, 0.434). C-to-base is positive throughout $(+0.50, +0.52, +0.27)$. S-to-base hovers near zero $(-0.06, -0.02, -0.09)$. Bootstrap CIs for S include zero at every layer; CIs for D and C both exclude zero. Cross-layer verdict: weak but robust (rank order $D \approx C > 0 > S$ preserved at every layer; CI separation suggestive but conservative under our both-sides bootstrap).

Reflection axis. All three trained conditions are strongly aligned with base on the reflection axis: D cosine 0.978, C 0.913, S 0.844 at L36, with comparable patterns at L24 and L28. D is the most

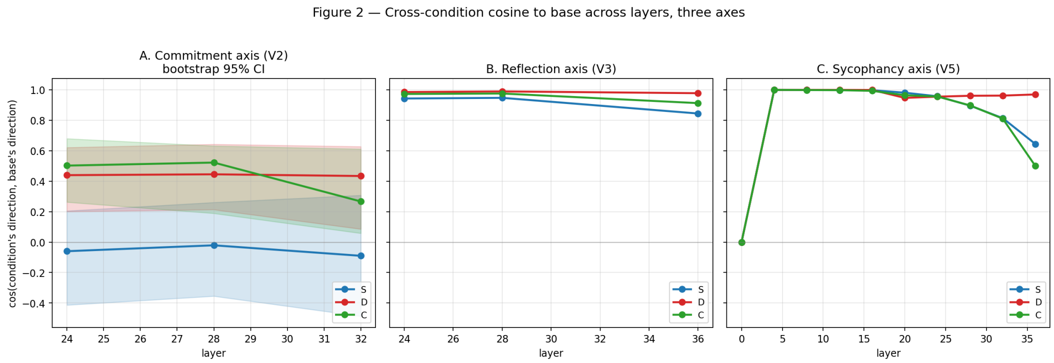

Figure 2: Cross-condition cosine to base across layers, three axes. **A:** Commitment direction with bootstrap 95% CI ribbons (layers L24/28/32). D $\approx$ C $> 0 >$ S at every layer probed. **B:** Reflection direction (layers L24/28/36). All three trained conditions strongly aligned with base (cosines 0.84–0.99); D most aligned. **C:** Peer-framing direction (all 10 probed layers). Conditions cluster tightly through mid layers; S diverges at L36.

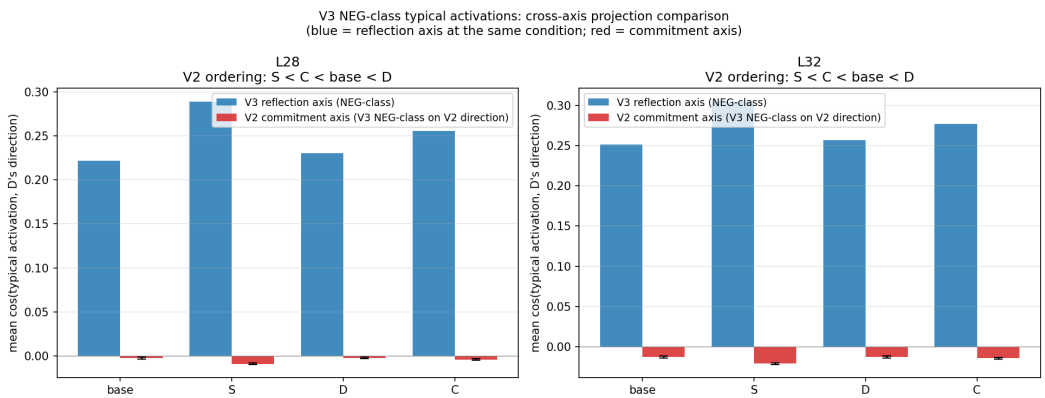

Figure 3: Typical-activation projection on the reflection (blue) and commitment (red) axes, using the same matched non-trigger token activations from the reflection-direction extraction as the "typical reasoning state" reference. On the reflection axis, the ordering is base $<$ D $<$ C $<$ S (S leans into the axis most, D sits at base-like engagement). On the commitment axis, the ordering inverts: S is pulled *away* from the axis, D sits at base-like engagement. Across both axes, D's typical state is the most base-like.

aligned with base at every layer. Direction norms across the four models are within 10% of each other (base 110.04, S 115.49, D 116.78, C 105.89 at L36), so training did not weaken D's reflection direction. Verdict: robust across layers.

Peer-framing axis. The four conditions are nearly indistinguishable in direction geometry through L32, with S diverging at L36. The interesting cross-condition signal on this axis is not in the direction geometry but in the typical-activation projection (§4.2).

4.2 The headline finding: D's typical state is base-like on every axis

The contrastive direction tells us whether each condition has a learned axis. The typical-projection analysis tells us how much each condition engages that axis during default reasoning, not at the contrastive boundary but in normal mid-rollout state. The two questions can dissociate, and in this work they do.

I project each condition's typical mid-reasoning activations (the matched non-trigger tokens used as the negative class of the reflection-direction contrast) onto D's reflection direction at L28 and L32, and separately onto D's commitment direction at the same layers. Figure 3 shows the result.

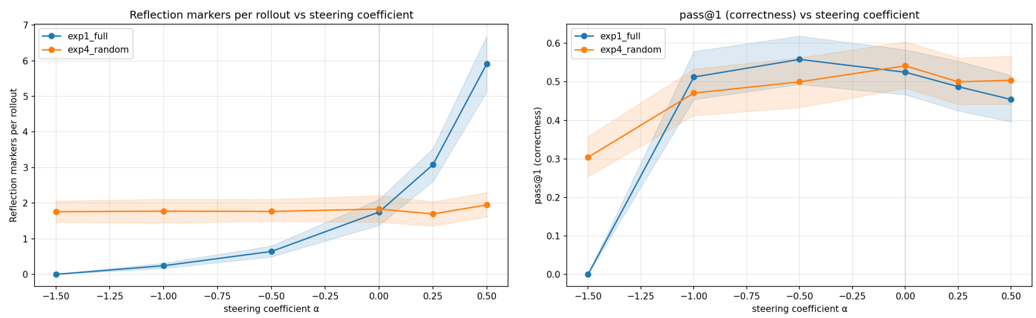

Figure 4: Reflection-direction steering on D at L28. **Left:** marker rate moves monotonically with α under the real direction (blue), from 0.24 markers/rollout at $\alpha{=}{-1}$ to 5.91 at $\alpha{=}{+0.5}$. Matched-norm random direction (orange) is essentially flat across the same range. **Right:** pass@1 stays in [0.45, 0.56] across the working range; only the most extreme suppression ($\alpha{=}{-1.5}$) collapses capability. Real and random tracks are similar on pass@1, so the reflection axis controls markers cleanly without dragging capability with it.

Reflection axis. Ordering at L28: base $+0.22 <$ D $+0.23 <$ C $+0.26 <$ S $+0.29$. S engages the reflection axis substantially more than D during typical reasoning. D sits at base-like engagement despite having the same contrastive norm as S. The reframe: D maintains the reflection axis as strongly as S, but D's default policy doesn't lean into that axis during normal generation. S's policy lives further along the reflection axis than D's.

Peer-framing axis (testing the peer-sycophancy hypothesis). The same diagnostic on the peer-framing direction. Ordering at L32: base $+0.097 <$ D $+0.104 <$ C $+0.118 <$ S $+0.126$. S is highest again; D sits barely above base. D's typical activations engage the peer-framing axis *less* than C's and substantially less than S's.

This directly falsifies an alternative explanation for D's adversarial robustness. The hypothesis was that D might be sycophantic toward peer input (since D was trained to integrate peer drafts) and that its resistance to wrong counter-arguments could come from peer-deference rather than genuine resistance. If that were true, D's typical processing should engage the peer-framing axis more than S's or C's. It engages it less. The peer-sycophancy explanation is rejected by this measurement.

Commitment axis. Same diagnostic on the commitment direction. Ordering at L32: S $-0.021 <$ C $-0.014 <$ base $-0.013 \approx$ D -0.013. The ordering inverts relative to the reflection axis: on reflection S is highest, on commitment S is lowest (most anti-aligned). D and base are essentially identical. Different mechanism per axis: solo training upregulates engagement on the reflection axis but pulls typical state *away* from the commitment axis. D preserves base-like engagement on both.

One feature unites D across all three axes: D's typical reasoning state sits at base-like engagement on every direction probed. Whatever distinguishes D behaviorally is not a continuous shift in default activation geometry. D's training affects what the model does at event moments (committing to an answer, deciding whether to emit a reflection trigger, processing a peer cue), not what it does throughout typical reasoning.

5 Causal validation via steering

Cross-condition geometry and typical-engagement projections are correlational. To convert these into causal claims I run three steering experiments. Each adds $\alpha \cdot direction$ to the residual stream output at a chosen layer during generation. Each includes a matched-norm random-direction control for specificity. Each includes an $\alpha = 0$ hook sanity check confirming the hook implementation is byte-identical to no-hook inference.

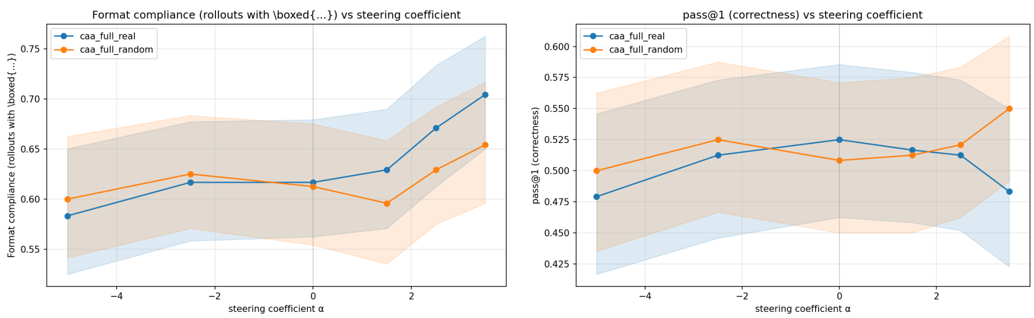

Figure 5: Commitment-direction steering on D at L32. **Left:** format compliance (rate of producing any `\boxed{...}`) climbs monotonically with α under the real direction; random is roughly flat. **Right:** pass@1 is flat-to-down across the entire α sweep under the real direction, with no advantage over random at any coefficient. The direction causally controls commitment behavior, not correctness.

5.1 Reflection-direction steering: clean causal control of surface markers

Setup. Steer D using D's reflection direction at L28, the layer where the direction is cleanest of unembedding-adjacency artifacts. Coefficients $\alpha \in \{-1.5, -1.0, -0.5, 0, +0.25, +0.5\}$. 30 MATH-500 L5 problems $\times$ 8 rollouts per (problem, α). Matched-norm random unit vector for the specificity control.

Marker rate moves monotonically with α; random control is flat. Figure 4 (left) shows the result. Under the real direction, marker rate moves from 0.24/rollout at $\alpha{=}-1.0$ to 1.75 at $\alpha{=}0$ to 5.91 at $\alpha{=}+0.5$, a roughly 24$\times$ swing. The random direction at matched norm stays at ≈ 1.8/rollout across the same α range. Specificity is confirmed: any L28 perturbation of this magnitude does not control markers; specifically perturbing along D's reflection direction does.

Capability is preserved across the working range. pass@1 stays in [0.454, 0.558] across $\alpha \in [-1, +0.5]$, tracking similarly between real and random. Format compliance also tracks similarly. $\alpha{=}-1.5$ collapses capability (format compliance drops to 0.004 and rollouts run to the 2048-token cap without emitting an answer). Past a threshold, suppressing the reflection axis breaks generation, but inside the working range steering reshapes surface behavior without dragging capability with it.

Transfer. Applying D's reflection direction to S at the same layer produces the same monotonic marker-rate shifts (S's marker rate moves from 0.46 at $\alpha{=}-1$ to 2.33 at $\alpha{=}0$ to 11.68 at $\alpha{=}+0.5$). The direction is not idiosyncratic to D; it acts as a control axis on S as well. This is also evidence that the axis I extracted is a real reflection direction shared across conditions, not a D-specific quirk.

5.2 Commitment-direction steering: it controls commitment, not correctness

Setup. Steer D using D's commitment direction at L32. Coefficients $\alpha \in \{-5, -2.5, 0, +1.5, +2.5, +3.5\}$; the commitment direction has smaller norm than the reflection direction so a wider α range is needed to produce comparable perturbation magnitudes. Otherwise identical setup.

The direction controls commitment behavior, not correctness. Figure 5 shows the headline. Format compliance climbs monotonically with α under the real direction: 0.583 at $\alpha{=}-5$, 0.617 at $\alpha{=}0$, 0.704 at $\alpha{=}+3.5$ (+12.1pp swing). Random direction at matched norm climbs only modestly. Revision rate (the rate of producing multiple distinct boxed values per rollout) spikes at $\alpha{=}+3.5$: 0.113 under real direction vs 0.021 under random. The direction has a clean specific causal handle on commitment-quality behavior.

But pass@1 is flat-to-down across the entire sweep. The real direction does not beat the matched random control at any α. pass@1 peaks at $\alpha{=}0$ (0.525) and drops at both extremes; at $\alpha{=}+3.5$ the

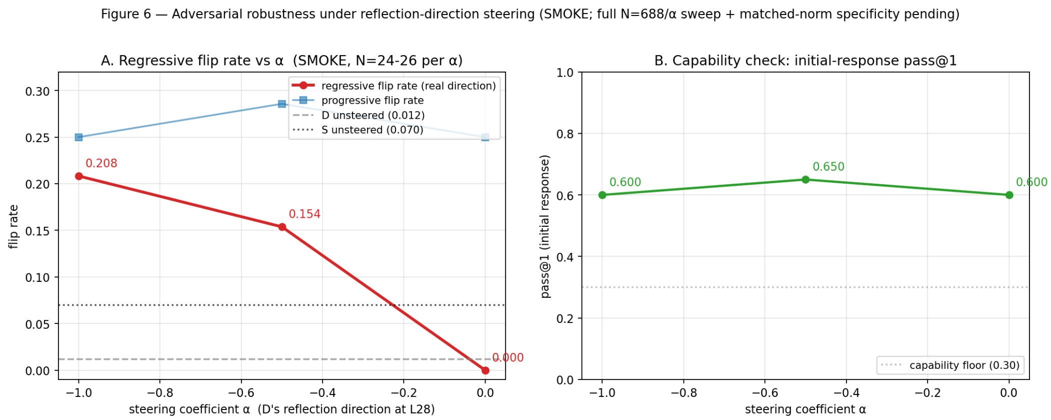

Figure 6: Adversarial robustness under reflection-direction steering. SMOKE result (5 problems $\times$ 8 rollouts $\times$ 3 α). **Left:** regressive flip rate (correct $\to$ wrong under wrong counter-argument) under real direction. Monotone dose-response: 0.000 at α=0, 0.154 at α=-0.5, 0.208 at α=-1.0. At α=-1.0 D is roughly 3$\times$ more vulnerable than unsteered S (dotted line at 0.070). Progressive flip rate (blue) flat across α. The reflection axis modulates resistance to wrong claims, not general persuadability. **Right:** initial-response pass@1 stays in [0.60, 0.65] across the sweep; capability preserved.

random direction is actually higher (0.550 vs 0.483). At no coefficient is there a directional advantage of the real direction over random on accuracy.

This is the reframe in a single experiment. The "correctness direction" extracted via CAA on rollout-level correct/wrong labels is causally an *answer-commitment direction*: pushing it positively makes the model commit more often (fmt up); pushing it too hard makes it commit and re-commit unproductively (revrate up, pass@1 down); but the pushed commitments do not preferentially land on correct answers.

The pass@1 correlation that the contrastive setup picks up in its train labels comes from the fact that correctly-answered rollouts commit cleanly to one boxed answer more often than incorrectly-answered ones; the direction reads that commitment-cleanliness signal, not correctness itself. This is consistent with the within-problem diagnostic being infeasible on our corpus. With no problems having both correct and incorrect rollouts at T=0.6, the contrast collapsed onto whatever is most discriminative between rollout-level labels at the model's commitment moment, which turns out to be commitment quality rather than correctness.

Methodological takeaway. For probing experiments that use rollout-level outcome labels with limited within-problem variance, the extracted direction may not be measuring what the label name suggests. Steering against a behavioral metric is a useful disambiguating check.

5.3 Adversarial robustness under reflection-direction steering (smoke)

The final causal claim is the most important one for the paper's headline story: does the reflection axis causally drive D's adversarial robustness, or is the robustness produced by something else?

Setup. Generate D's initial answer under steering, then construct a counter-argument (regressive if initially correct, progressive if wrong), then generate D's reconsidered answer with the hook still active. $\alpha \in \{0, -0.5, -1.0\}$. The smoke runs 5 problems $\times$ 8 rollouts per α, giving N≈24-26 in the regressive denominator per α (other rollouts produce wrong initial answers and go into the progressive denominator).

Suppressing the reflection axis makes D adversarially vulnerable. Figure 6 shows the result. Regressive flip rate moves from 0.000 at α=0 (matching D's unsteered baseline of 0.012) to 0.154 at α=-0.5 to 0.208 at α=-1.0. At α=-1.0 — a coefficient that elsewhere leaves capability intact —

D is roughly 3$\times$ more vulnerable to wrong counter-arguments than unsteered S (S=0.070). Initial-response pass@1 stays in [0.60, 0.65], so capability is preserved across the sweep; the increased flipping is not a side effect of D becoming generally weaker.

Progressive flip rate (accepting a correct counter-argument when initially wrong) is roughly flat across α at ≈ 0.25. The reflection axis modulates resistance to wrong claims, not general persuadability. This asymmetry matches the asymmetry in D's behavioral fingerprint: D is dramatically more resistant to wrong counter-arguments than S without being correspondingly more resistant to correct ones.

Caveat. This is a smoke result. The full sweep on the 86-problem intersection and the matched-norm random-direction specificity control are both in progress; until the specificity check lands, the smoke result is consistent with "any L28 perturbation makes D adversarially vulnerable" rather than the more specific "the reflection direction specifically does." The marker-rate specificity check (Figure 4) is indirect evidence that L28 perturbations along arbitrary directions don't produce the same behavioral effects as steering along the reflection direction, but the direct check on adversarial flipping is what closes the claim.

6 What discussion teaches

The three axes converge on one finding: D's training preserves base-like *typical* representational state on every axis measured, while sharpening the discrimination D's representation makes *at event moments*. Most cleanly at counter-argument encounters, where suppressing the reflection axis specifically eliminates D's resistance.

This answers the question the project started with. Discussion teaches event-time discrimination on the reflection axis. It does not teach the model to reflect more during default reasoning (D's typical state is base-like, not elevated like S's). It does not teach the model to peer-defer (D engages the peer-framing axis the least of the trained conditions). What it teaches is something that fires specifically when the model is faced with a counter-argument: at those moments, the reflection axis is causally responsible for keeping D on its correct initial answer.

Why doesn't this translate to a pass@1 win over C? The persuader bonus D's training is supposed to provide is heavily diluted by the symmetry of the two players: about 47% of disagreement outcomes are mutual swaps where both players flip rather than one persuading the other. The signal is real but sparse. The natural fix is training D with a heterogeneous Bob (different model or checkpoint, so the two players have uncorrelated failure modes), which is what I'm working on for the follow-up.

7 Limitations and next steps

Mutual swap dominates D's training signal. Only $\sim$15% of disagreement outcomes are productive asymmetric persuasion; $\sim$47% are mutual swaps. The cleanest test of whether discussion can outperform a compute-matched control is to break this symmetry. The plan is a heterogeneous-Bob version of D (D-prime) where Bob is sampled from a different model or checkpoint so the two players have uncorrelated failure modes. If D-prime produces a higher productive-persuasion rate and beats C on hard OOD evals, the discussion mechanism is doing real work; if not, the negative accuracy result generalizes beyond the symmetric-player setup. This is the immediate follow-up.

Adversarial-under-steering needs the full sweep and specificity control. The smoke result is unambiguous in direction and magnitude but uses only 5 problems. The full 86-problem sweep is queued, as is the matched-norm random-direction control. Until the specificity check lands, the smoke is consistent with "any L28 perturbation makes D adversarially vulnerable." The marker-rate specificity result is indirect evidence for the more specific claim but is not a substitute for the direct check.

Commitment direction's within-problem diagnostic is infeasible. The standard sanity check for whether CAA captures the labeled construct rather than a correlate is the within-problem-vs-corpus comparison. With 8 rollouts per problem at T=0.6, no problems in our corpus have both correct and incorrect rollouts, so the within-problem analysis can't be run. The steering experiment substitutes for this missing geometric check and is what produced the commitment-vs-correctness reframe.

MATH-500 L5 pending re-evaluation. A preliminary pass@k sweep on MATH-500 L5 (n=134, 256 rollouts per problem) suggested the instruct base outperforms all three trained conditions on this benchmark, which would contradict the C $\geq$ D $>$ S $>$ base ordering on L4 and GPQA. The result is surprising enough that I'm treating it as unconfirmed: it may be an evaluation pipeline bug or a benchmark contamination artifact in the instruct base. The benchmark setup is being re-checked before being reported.

Single base model, single seed, single discussion design. All four models are Qwen3-4B variants and each condition was trained once. The negative accuracy result against C is specific to the symmetric-player protocol tested. Multi-seed runs, a different base model, and alternative protocols (heterogeneous Bob, stability bonus, curriculum widening) are all plausible follow-up work; heterogeneous Bob is the most direct test of whether the discussion-vs-compute-control finding generalizes.

8 Related work

Multi-agent RLVR training. MAPoRL [Park et al., 2025] co-trains multiple LLMs with discussion via RL. CRISP [Baumgartner and Agrawal, 2026] introduces a discussion-on-disagreement protocol with coach-regulated difficulty. Dr. MAS [Feng et al., 2026] proposes agent-wise advantage normalization for stable multi-agent RL. All three analyze behavioral outcomes; none isolate discussion from compute, and none probe internal representations.

Inference-time multi-agent comparisons. Wang et al. [2024] finds Self-Consistency matches or beats Multi-Agent Debate at matched compute on most tasks. Becker et al. [2026] argues a Data Processing Inequality bound on multi-agent vs matched single-agent systems. These are inference-time results on frozen models; we ask the analogous question during training.

Linear representations and CAA. Marks and Tegmark [2024] establishes that high-level concepts are encoded as linear directions in residual streams. Rimsky et al. [2024] introduces the contrastive activation addition methodology we use for both direction extraction and steering. Arditi et al. [2024] applies related techniques to behavioral steering. Zhu et al. [2025] extracts a self-reflection vector via a contrastive setup we adapt for the reflection axis.

Mechanistic analysis of training. Betley et al. [2026] shows narrow fine-tuning induces broad misalignment; Chen et al. [2025] shows persona vectors partially explain such training-induced changes. Our methodological template is similar: train matched models that differ in one variable, then probe internal representations.

9 Conclusion

At matched compute, discussion training and additional solo sampling produce models that are close in accuracy but distinct in behavior. D's adversarial flip rate is roughly 6$\times$ lower than S's and 4$\times$ lower than C's. The mechanism is not a continuous shift in default reasoning state: D's typical activations engage the reflection, commitment, and peer-framing axes at base-like levels. What D's training does is sharpen event-time discrimination on the reflection axis, and the smoke result shows this axis is causally responsible for D's adversarial robustness.

The pass@1 comparison between D and C does not favor D under the symmetric-player protocol I tested. Mutual swap dilutes the discussion training signal; breaking player symmetry is the immediate next step. Whether a heterogeneous-Bob D translates the robustness fingerprint into accuracy gains is the open question that follows.

References

A. Arditi, O. Obeso, A. Syed, et al. Refusal in language models is mediated by a single direction. In *NeurIPS*, 2024.

A. Baumgartner and S. Agrawal. CRISP: Coach-regulated iterative solver pairing for mathematical problem solving. Course report, University of Chicago, 2026.

J. Becker et al. SAS vs MAS: A Data Processing Inequality view of multi-agent systems at matched compute. 2026.

J. Betley, N. Warncke, A. Sztyber-Betley, et al. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649(8097):584–589, 2026.

R. Chen, A. Arditi, H. Sleight, O. Evans, and J. Lindsey. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*, 2025.

L. Feng, L. Zheng, S. He, F. Zhang, and B. An. Dr. MAS: Stable reinforcement learning for multi-agent LLM systems. *arXiv preprint arXiv:2602.08847*, 2026.

D. Guo, D. Yang, H. Zhang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

D. Hendrycks, C. Burns, S. Kadavath, et al. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS*, 2021.

S. Marks and M. Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *COLM*, 2024.

C. Park, S. Han, X. Guo, et al. MAPoRL: Multi-agent post-co-training for collaborative large language models with reinforcement learning. In *ACL*, 2025.

D. Rein, B. L. Hou, A. C. Stickland, et al. GPQA: A graduate-level Google-proof Q&A benchmark. In *COLM*, 2024.

N. Rimsky, N. Gabrieli, J. Schulz, et al. Steering Llama 2 via contrastive activation addition. In *ACL*, 2024.

H. Wang et al. Rethinking the bounds of LLM reasoning: Are multi-agent debates the right path? In *NeurIPS Workshop*, 2024.

Q. Yu, Z. Zhang, R. Zhu, et al. DAPO: An open-source LLM reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.

J. Zhang et al. Reasoning models know when they're right: Probing intermediate answer correctness. *arXiv preprint arXiv:2504.05419*, 2025.

Z. Zhu et al. The reasoning vector: A self-reflection direction in RLVR-trained models. *arXiv preprint arXiv:2506.12217*, 2025.

A. Zou, L. Phan, S. Chen, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Hyperparameters and training details

Table 1 lists training hyperparameters shared across all three conditions, with footnotes for condition-specific differences.

Persuader bonus formula (D only). A player p is the *persuader* for a given problem if: (i) the majority pre-discussion answer of p is correct, (ii) the majority pre-discussion answer of the peer $\neg p$ is incorrect, and (iii) the majority post-discussion answer of $\neg p$ is correct. When these three conditions all fire, every correct pre-discussion rollout of p receives an additive bonus $\gamma_1 = 0.3$ on top of its base task reward. No bonus is applied in any other case (mutual swap, both correct, both wrong, or peer didn't flip). The bonus is additive, not multiplicative.

Compute-matching protocol for C (self-draft). On each disagreed problem (at least 2 distinct boxed answers among parseable rollouts), each player generates one additional rollout from a self-draft prompt structured identically to D's discussion prompt, except the two solutions shown are both from the same player (selected via preference for both-correct or both-wrong pairs). The extra rollout joins the player's training pool for that problem and receives standard GRPO advantage and gradient treatment. The compute spent matches D's discussion-stage compute exactly: 1 extra rollout per player per disagreed problem, same prompt structure, no peer signal.

Table 1: Training hyperparameters. Phase 1 matched across S/D/C; footnotes mark condition-specific deviations.

Parameter	Value
Base model	Qwen3-4B-Instruct-2507 (36 layers, hidden 2560, bf16)
GRPO algorithm	DCPO-style asymmetric-clip GRPO
Rollouts per problem per player	8
Problems per batch	8
Optimizer	DeepSpeed FusedAdam (β_1=0.9, β_2=0.999), wd 0.01
Peak learning rate	1×10^{-6}
LR warmup	Linear, 20 iters
LR decay (post-warmup)	Cosine to $0.1\times$ peak by iter 200
Clip ratio ϵ (low / high)	0.2 / 0.28
DCPO α (importance-ratio scaling)	3.0
KL coefficient (forward JS vs frozen base)	0.005
Sampling temperature, Alice	0.6
Sampling temperature, Bob (D, C only)	1.0
Max sequence length	12,288 tokens
Training data	DAPO-Math-17K (HuggingFace `open-r1/DAPO-Math-17k-Processed`)
Curriculum filter	2,023 problems, solve-rate band $[0.25, 0.75]$ at T=0.6
Total iterations	200
Hardware	2× H100, DeepSpeed ZeRO-2, AdamW offload
Task reward	SymPy-verified binary correctness on final `\boxed{}`
No-box penalty	-0.5
Persuader bonus γ_1 (D only)	$+0.3$ additive
Compute-control mode (C only)	`self_draft`, 1 extra rollout per player per disagreed problem

B Supplementary figures

B.1 Reflection axis

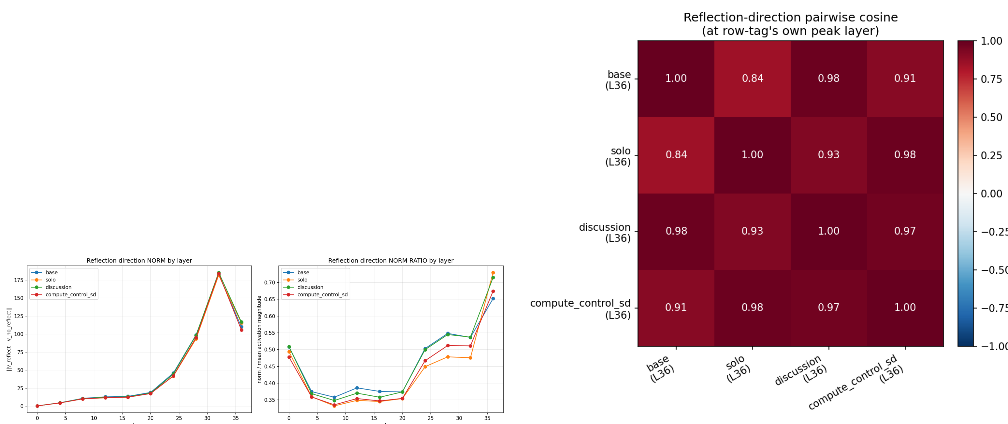

Figure 7: Reflection direction. **Left (S1):** L2 norm by layer for all four conditions. Norm grows monotonically and peaks at L36 with values within 10% of each other (base 110.0, S 115.5, D 116.8, C 105.9). **Right (S2):** pairwise cosine of the reflection direction at each model's L36 peak. All trained conditions remain highly aligned with base (S 0.844, D 0.978, C 0.913) and with each other (0.91–0.98). Reflection direction preserves orientation across training; the cross-condition difference lives in typical engagement (Figure 3), not in direction geometry.

B.2 Peer-framing axis

B.3 Commitment axis

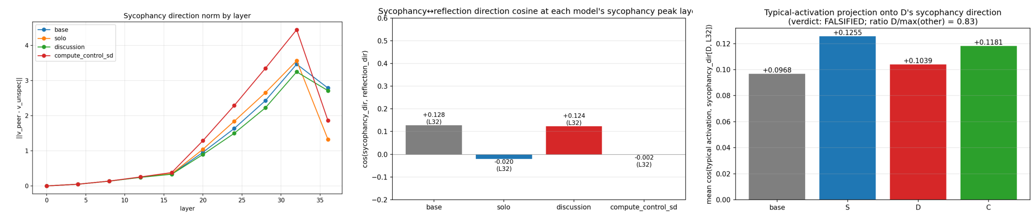

Figure 8: Peer-framing axis. **Left (S3):** direction norm by layer. L0 norm is exactly zero by construction (matched-token boundaries), a hard sanity check. Peak at L32: base 3.47, S 3.57, D 3.25, C 4.45. D has the *smallest* norm of the four conditions, opposite the prediction of the peer-sycophancy hypothesis. **Center (S4):** cosine between each condition's peer-framing and reflection directions at L32 (base $+0.128$, S -0.020, D $+0.124$, C -0.002). All near zero; the two axes are essentially independent. **Right (S5):** typical-activation projection onto D's L32 peer-framing direction. Base $+0.097$, D $+0.104$, C $+0.118$, S $+0.126$. D engages this axis less than C or S, falsifying the peer-sycophancy hypothesis ($D/\max = 0.83$, below the supported threshold).

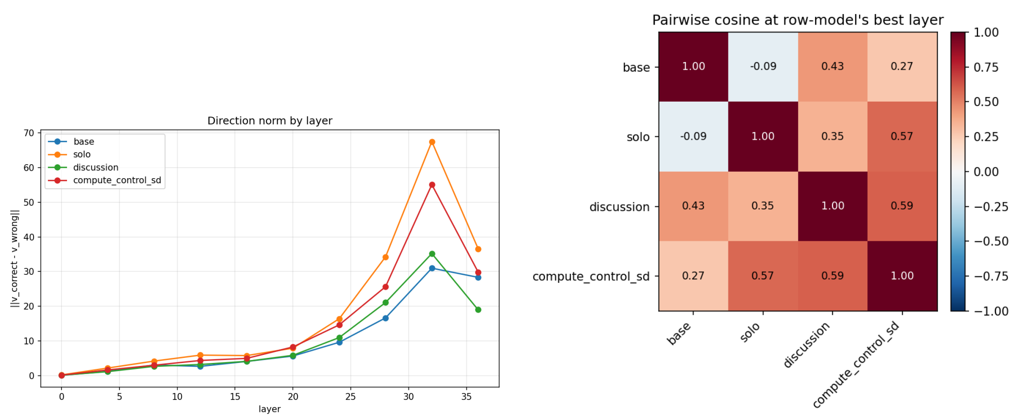

Figure 9: Commitment direction. **Left (S6):** direction norm by layer. Peak at L32: base 30.99, S 67.52, D 35.21, C 55.10. S has the largest commitment direction (norm-ratio 0.207 vs base 0.096, D 0.111, C 0.171), consistent with strong gradient pressure to distinguish correct from wrong commits during solo RL. D's norm is base-like. **Right (S7):** pairwise cosine at L32. D-to-base $+0.434$ and C-to-base $+0.267$ are positive; S-to-base is -0.090. S sits in a different commitment-axis orientation than base while D and C preserve alignment. This is the geometric signature behind the cross-condition pattern in Figure 2A.